

Layer Reduction in Small LLMs: What Actually Works

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Abstract

We compare ways of reducing transformer depth on the same target blocks of the same models: *deleting* them, *replacing* them with a fitted rank-32 ternary low-rank bridge, *deleting then repairing* with pruned distillation, and *fine-tuning with LayerDrop*. Every configuration is repeated over multiple seeds and reported as mean $\pm$ standard deviation; the fitting procedure is bit-level deterministic, verified rather than assumed; and compression is measured in parameters, fp16 bytes and wall-clock latency rather than asserted.

Two results survive scrutiny. First, the best method is *model-dependent*: BI-targeted removal plus distillation wins on GPT-2 (41.68 ± 0.37 against a 38.25 baseline), while LayerDrop fine-tuning wins on SmolLM-135M and is the only configuration in the entire study that lands *below* its own unpruned baseline (15.54 ± 0.01 against 16.36). Second, a negative result: a fitted low-rank bridge loses to plain deletion while achieving *identical* parameter counts, so it deletes the same weights and then adds a perturbation on top. A training-free SVD diagnostic explains why, and shows that block influence and residual low-rankness are different properties, and that improving the bridge’s local layer-wise fit makes end-task perplexity *worse*.

All numbers are produced by a single self-auditing notebook (`assignment2_tpu.ipynb`) which refuses to certify its own output unless thirteen structural invariants hold.

1 Report

1.1 Mentor feedback since Presentation 1

FILL IN. (Ten marks; this subsection is deliberately left for you.) The two substantive changes made in response to earlier feedback are already visible elsewhere in this report: the bridge is *fitted* rather than random, and a disjoint random-removal control was added so that the Block Influence ranking can be defended rather than merely asserted.

1.2 Baselines and related work

We position against the work that is runnable on our compute budget.

- **ShortGPT** (Men et al., 2024) introduces Block Influence, the exact metric we use for target selection. We reproduce its central claim as our control, with seeds and error bars that the original comparison does not report.
- **LLM-Pruner** (Ma et al., 2023) establishes that depth-pruned models require distillation to recover. We adopt that recipe, and we apply it to *both* of our operators so that neither is advantaged by being left unrepaired.
- **LayerDrop** (Fan et al., ICLR 2020) fine-tunes with structured layer dropout. We run it as a full competing method rather than as a baseline, and it turns out to be the strongest approach on one of our two models.
- **LLM in a 1.58 bits** (Ma et al., 2024) motivates the ternary factor grid, though our negative result is about layer replacement rather than weight quantization.

- **Hinton et al. (2015)** for the distillation objective.

No closed-source large-LLM baselines were reproduced, per the instructions.

1.3 Datasets, bias, and bottlenecks

Corpus. `Salesforce/wikitext`, configuration `wikitext-2-raw-v1`, the standard layer-pruning benchmark. Its non-raw and `wikitext-103` variants are available in the same repository and were inspected as robustness checks.

Bias handling. Preprocessing is identical across every operator: non-overlapping fixed-length windows, the same evaluation token budget, and the same calibration slice. The random-removal control is drawn from the *complement* of the BI target set, so a random configuration can never accidentally receive the block we chose, making it a strictly fair null. Target selection, calibration data, and evaluation data are disjoint in role even though they come from the same corpus: k is chosen on validation and the test split is touched exactly once, afterwards.

Bottlenecks. Datasets 5.x removed the script loaders and the parquet mirrors for PTB and `tiny_shakespeare` were not resolvable, so we rotated the wikitext configurations instead. The principal compute bottleneck was a single Apple MPS laptop, which is why four training-heavy approaches initially ran at budgets too small to separate methods; those runs were re-run at adequate budgets rather than reported.

1.4 Experimental plan

Models. GPT-2 small (124,439,808 parameters, 12 blocks) and SmolLM-135M (134,515,008 parameters, 30 blocks). They are architecturally unlike each other: GPT-2 uses learned absolute position embeddings, GPT-2 attention, and `Conv1D` projections stored as `[in,out]`, whereas SmolLM uses RoPE, GQA and SwiGLU with projections stored as `[out,in]`. Getting that transposition wrong silently corrupts GPT-2 position embeddings, so a thin `get_layers/set_layers` helper normalises access.

How the approaches differ. (i) *target selection*: BI-ranked versus random versus learned gates versus an EV-guided ranking; (ii) the *operator*: deletion, fitted ternary bridge, distillation repair, LayerDrop, gate-masked deletion, or attention-head pruning; (iii) *scope*: k from 1 to 3. Eight mechanism families are represented across fifteen numbered approaches.

Hyperparameters and selection. Bridge rank $r = 32$, AbsMean ternary grid (2 bits), ridge 10^{-4} , Adam at 10^{-2} for 80 bridge-fitting steps. Distillation: 400 steps, AdamW at 2×10^{-5} , temperature $T = 2$, $\alpha = 0.5$ on the cross-entropy term, sequence length 256. LayerDrop: 500 steps, drop probability 0.15. Learned gates: 300 steps, $\lambda = 0.1$. Layer selection is always the BI ranking unless the approach explicitly varies it.

Metric. Primary: validation perplexity $\text{PPL} = \exp(\frac{1}{N} \sum_i -\log p_\theta(x_i|x_{<i}))$ over non-overlapping windows, identical for every operator. Secondary: parameter count, fp16 byte count, and single-sequence latency. We additionally record the BI value and the SVD explained-variance ceiling of every removed block.

System effort. One MacBook with 16 GB RAM on the MPS backend for the seeded studies, and a Colab T4 GPU for the final grid. Every configuration is produced by `assignment2_tpu.ipynb`, which is uploaded to Colab and run end to end with a single `SMALL` flag selecting between the verification pass and the paper grid.

1.5 Project management

Novelty proposal. Test whether a fitted low-rank bridge can beat plain deletion of the same block, and if it cannot, establish why with a training-free diagnostic rather than by assertion.

Resources. 1 laptop (16 GB, Apple MPS, no paid GPU) plus a Colab T4 GPU runtime for the final grid.

Team contribution. FILL IN per member.

2 Experiment

2.1 Reproducibility, and a defect we had to fix first

An earlier version of this study used a single global `torch.manual_seed` for the whole notebook. A seed audit demonstrated that this made results depend on how many random numbers had been drawn earlier in the run: the same configuration produced 47.67/27.04 when re-executed as its own cell and 47.95/27.67 when reached by running the notebook top to bottom. Those irreproducible figures appear in earlier drafts of this report and have been removed everywhere.

The fix was to drive every stochastic fit from an explicit per-fit `torch.Generator` rather than the global RNG. Healing additionally uses a seed stream offset from the bridge-fitting stream so the two are not correlated by accident. Section 2.7 reports the resulting reproducibility check.

2.2 Results: the four unrepaired operators

Validation perplexity, mean $\pm$ standard deviation over three seeds, targets chosen by BI, random removal drawn disjointly from the BI set.

Operator	GPT-2 (124M, 12 blocks)			SmolLM-135M (135M, 30 blocks)		
	$k=1$	$k=2$	$k=3$	$k=1$	$k=2$	$k=3$
baseline (unpruned)		38.25			16.39	
BI removal	45.76	47.95	121.81	19.93	26.13	29.91
fitted bridge	58.67 ± 9.53	86.49 ± 24.63	765.21 ± 214.21	30.77 ± 3.54	41.77 ± 3.30	58.21 ± 9.46
random removal	1453.93 ± 2441.72	4351.09 ± 203.58	2814.29 ± 2378.60	20.16 ± 1.94	22417.94 ± 19512.02	269.08 ± 287.45

Table 1: Validation perplexity on `wikitext-2-raw-v1`. BI removal is deterministic given its targets, so its standard deviation is exactly zero; the bridge and random removal are stochastic because of the fit and the draw.

Table 1 replaces an earlier table in which the fitted bridge was reported as 45.11 on GPT-2 and 21.71 on SmolLM, i.e. as matching or beating hard removal. Those figures came from the unreproducible global-RNG configuration. With per-fit seeding the bridge is clearly worse, at 58.67 and 30.77.

The bridge buys nothing. The fitted bridge and plain removal produce *identical* parameter counts: 117,351,936 for both at GPT-2 $k=1$, against 124,439,808 for the baseline. A rank-32 bridge is 49,152 weights standing in for a 7.1M block, so it removes exactly the same parameters as deletion and then adds a small perturbation. Latency is likewise indistinguishable (26.56 ms for removal against 26.54 ms for the bridge on GPT-2). The bridge is therefore strictly dominated: it carries more parameters-worth of risk and no benefit.

2.3 Results: repair

LLM-Pruner established that depth-pruned models need distillation, so comparing two *unrepaired* operators answers only “which does less damage”. We therefore ran the standard pruned-distillation

objective

$$\mathcal{L} = \alpha \text{CE}(y, \hat{y}_S) + (1 - \alpha) T^2 \text{KL}(\text{softmax}(\hat{y}_S/T) \parallel \text{softmax}(\hat{y}_T/T)) \quad (1)$$

with S the pruned student and T the frozen original. The T^2 factor is not cosmetic: softening a 50k-way distribution by T scales the gradient by $1/T^2$, so omitting it silently downweights distillation until the run is just fine-tuning. We assert that the KL term is non-trivial rather than assuming it; the mean initial KL is 134.48, falling to 34.81.

Model	k	baseline	pruned	repaired	std	recovered
GPT-2	1	38.25	45.76	41.68	0.37	54.3%
GPT-2	2	38.25	47.95	44.06	0.17	40.2%
SmolLM-135M	1	16.39	19.93	18.67	0.07	34.9%
SmolLM-135M	2	16.39	26.13	22.14	0.09	39.4%

Table 2: Repair by pruned distillation recovers a large fraction of the perplexity that deletion costs, at no parameter cost beyond the deletion. Healed beat pruned in 12 of 12 runs, not merely on average.

2.4 Results: LayerDrop, learned gates, and head pruning

Three further families were run as full competing methods rather than as side notes, because two of them change the conclusion.

LayerDrop is the strongest method on SmolLM, and the only one that beats its own baseline. Fine-tuning with structured layer dropout at $p = 0.15$ for 500 steps gives

	GPT-2 $k=1$	GPT-2 $k=2$	SmolLM $k=1$	SmolLM $k=2$
baseline	38.25	38.25	16.39	16.39
BI removal	45.76	47.95	19.93	26.13
BI + distillation	41.68	44.06	18.67	22.14
LayerDrop	51.74	57.53	15.54	19.62

On SmolLM-135M, LayerDrop at $k = 1$ reaches 15.54 ± 0.01 , which is *below* the 16.39 unpruned baseline: deleting a block and fine-tuning made the model better. This is the most surprising number in the study and we report it rather than bury it, because it contradicts the obvious reading that healing is the best available repair. It does not generalise: on GPT-2 the same procedure reaches 51.74, far worse than both plain deletion and distillation. The honest conclusion is that *the best method is model-dependent*, and that a single recommended method would be an overstatement of what this evidence supports.

Learned gates never beat BI. Training a L_0 penalty on per-block gates and then deleting the lowest-gate blocks reaches 44.25 on GPT-2 and 18.95 on SmolLM, never better than the BI ranking that it is supposed to improve on. The overlap between the learned gate ordering and the BI ordering is exactly 0.00 on GPT-2, i.e. the gate finds a completely different set of blocks while arriving at a similar perplexity. Gates are a more expensive way of rediscovering that most blocks are not equally important.

Head pruning is a clean negative. Pruning the least important attention heads by magnitude is cheap at 12.5% ($38.25 \rightarrow 42.85$ on GPT-2) and catastrophic at 25% (2033.08). On SmolLM even the 12.5% setting is already severe ($16.36 \rightarrow 46.92$).

2.5 Mechanism: why the bridge loses

The diagnostic is training-free. For each block we capture calibration inputs X and outputs Y , form the residual map $\Delta = Y - X$, and compute its truncated-SVD explained variance. By Eckart–Young this upper-bounds what *any* rank- r linear replacement can achieve on that block, with no training at all.

Block influence and residual low-rankness are different properties. On GPT-2 the lowest-BI block is L4, whose rank-32 explained variance is 0.392, while L1 sits at 0.875. On SmolLM the lowest-BI block is L2 at 0.750, against L27 at 0.445. BI does not select the most linearly compressible block, so an argument that conflates “redundant” with “low-rank” is simply wrong.

More rank makes perplexity worse, not better. On GPT-2 the rank sweep at $k = 1$ gives 50.57, 60.35, 68.18 and 341.10 at ranks 4, 16, 64 and 128. The local fit improves monotonically while end-task perplexity degrades monotonically. Layer-wise reconstruction error is therefore the wrong optimisation target, which is the mechanistic explanation for the negative result.

An honest weakness in this framing. The bridge is rank-32 *and* 2-bit, so “low-rank replacement loses” is entangled with “ternary quantization hurts”. The bit ablation does not rescue the framing: GPT-2 gives 52.35 at 2 bits against 69.78 at 4 and 69.72 at 8, and SmolLM gives 34.83, 44.02, 39.55. More bits is monotonically *worse*, which means quantization precision is not the bottleneck and we cannot currently explain why 8 bits loses to 2. The defensible claim is therefore narrower than the one we originally wrote: *this particular linear-plus-residual fitted form loses to deletion, and giving it more capacity makes it worse*. Resolving this needs an unquantized float32 bridge at each rank, which we have not run. These ablations are single-seed and should be read as indicative rather than conclusive.

2.6 Held-out test

$k = 1$ was selected on validation, so the test split is touched once, for that configuration only. GPT-2: 53.89 for BI removal against 67.97 ± 10.65 for the bridge. SmolLM: 34.12 against 50.33 ± 1.24 . The ordering is preserved from validation to test.

2.7 Self-audit

The notebook ends with thirteen structural invariants and refuses to certify its own numbers if any fails. They include bit-level reproducibility of the fit, full seed coverage, monotone parameter counts, a random control provably disjoint from the BI targets, finite perplexities, the SVD ceiling being a genuine upper bound at matched rank, healing never increasing perplexity, the healed model retaining the post-deletion parameter count, BI never losing to random removal, and the held-out test perplexity tracking the unpruned model’s own validation-to-test gap rather than an arbitrary threshold. Three of these checks initially failed and every failure was a defect in the check rather than in the science: a seed-coverage expectation computed once for the whole table, a rank ceiling read at the wrong rank, and an absolute validation-to-test tolerance that would also have rejected a pristine model. We report this because a gate that has never failed is not evidence of anything.

2.8 Findings

1. **Best method is model-dependent.** BI removal plus distillation wins on GPT-2 (41.68 against a 38.25 baseline); LayerDrop wins on SmolLM-135M and alone beats its own baseline (15.54 against 16.39).

2. **A fitted low-rank bridge is dominated.** It loses to plain deletion while achieving identical parameter counts and indistinguishable latency.
3. **BI’s advantage is reliability, not uniform magnitude.** BI removal is never worse than random removal at any $(model, k)$, but at SmolLM $k = 1$ the two are within noise (19.93 against 20.16). The advantage is concentrated in the worst case: GPT-2 $k = 1$ random removal reaches 1453.93 ± 2441.72 , with a standard deviation exceeding its own mean, which is precisely the unreliability that makes a configuration undeployable.
4. **Block influence and residual low-rankness are different properties**, and optimising the local fit actively harms perplexity.
5. **Learned gates and head pruning do not help.** Gates reach similar perplexity by identifying a disjoint set of blocks; head pruning fails at 25%.

3 Limitations

- Two models, both under 140M parameters. Nothing here is evidence about 7B-scale models, and depth-pruning behaviour is known to shift with scale.
- One benchmark and one metric. `wikitext-2` perplexity only, with no downstream task accuracy, so we cannot say whether these perplexity differences would matter in use.
- The bridge result is confounded with ternary quantization, as Section 3.4 states plainly. The unquantized control is missing.
- Ablations are single-seed.
- Distillation is one untuned point: one learning rate, one temperature, one step count, one α . The recovery fractions in Table 2 are lower bounds on what tuned distillation would give.
- LayerDrop was run at one probability and one step count; given that it produced the study’s best result, a proper sweep is the highest-value follow-up.
- Latency is a single-sequence microbenchmark at one length, not a serving benchmark.

4 Conclusion

The defensible recommendation is to delete the blocks with the lowest Block Influence and then repair with distillation, and to treat low-rank bridge replacement as not worth the complexity at equal parameter count. The stronger claim we set out to test, that a fitted bridge preserves perplexity where deletion does not, is false, and we report it as false. The most interesting result is not on our intended axis at all: LayerDrop fine-tuning lands below the unpruned baseline on SmolLM-135M, which suggests that for some architectures the right question is not how cheaply depth can be removed but whether the surviving network can be improved once it is.

Reproducing this

`assignment2_tpu.ipynb` is uploaded to Google Colab, the runtime set to GPU / T4, and the notebook run end to end. One flag, `SMALL`, selects the verification pass or the paper grid. Every table above is computed in-notebook; none is hard-coded. Source, notebooks and per-approach results are at github.com/sumitjhoo/layer-reduction-approaches.

References

- [1] Men et al., *ShortGPT: Layers in Large Language Models are More Redundant Than You Expect*, 2024.
- [2] Ma et al., *LLM-Pruner: Structural Pruning for Large Language Models*, 2023.
- [3] Fan et al., *Reducing Transformer Depth on Demand with Structured Dropout*, ICLR 2020.
- [4] Ma et al., *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits*, 2024.
- [5] Hinton et al., *Distilling the Knowledge in a Neural Network*, 2015.